Coordinate Component Methods for Computing Curvature

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Summary

This note provides a self-contained derivation of the Riemann curvature tensor and the Ricci tensor using the coordinate component method, based on the definition of the Riemann tensor and the properties of the Christoffel symbols. Explicit component formulas are given, and an application to the divergence of a vector field is also included.

Keywords: Riemann curvature tensor, Ricci tensor, Christoffel symbol, coordinate component method

The Riemann curvature tensor satisfies

$$R_{abc}{}^{d}\omega_d = \nabla_a \nabla_b \omega_c - \nabla_b \nabla_a \omega_c, \tag{1}$$

and the Ricci tensor

$$R_{ac} = R_{abc}{}^{b} \tag{2}$$

is obtained by contracting the second and fourth indices of the Riemann curvature tensor. In this section, we shall compute the explicit expression for $R_{abc}{}^{d}$ using the coordinate component method, based on Eqs. (1) and (2).

1 The Riemann Curvature Tensor

From the formula

$$\nabla_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} = \tilde{\nabla}_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} + \sum_i C^{b_i}{}_{ad} T^{b_1\cdots d\cdots b_k}{}_{c_1\cdots c_l} - \sum_j C^{d}{}_{ac_j} T^{b_1\cdots b_k}{}_{c_1\cdots d\cdots c_l}, \tag{3}$$

it follows that

$$\nabla_b \omega_c = \partial_b \omega_c - \Gamma^{d}{}_{bc}\omega_d, \tag{4}$$

and thus,

$$\nabla_a \nabla_b \omega_c = \partial_a(\partial_b \omega_c - \Gamma^{d}{}_{bc}\omega_d) - \Gamma^{e}{}_{ab}(\partial_e \omega_c - \Gamma^{d}{}_{ec}\omega_d) - \Gamma^{e}{}_{ac}(\partial_b \omega_e - \Gamma^{d}{}_{be}\omega_d). \tag{5}$$

Hence

$$R_{abc}{}^{d}\omega_d = \nabla_a \nabla_b \omega_c - \nabla_b \nabla_a \omega_c$$

$$
\begin{aligned}
&= \left[\partial_a(\partial_b\omega_c - \Gamma^d{}_{bc}\omega_d) - \Gamma^e{}_{ab}(\partial_e\omega_c - \Gamma^d{}_{ec}\omega_d) - \Gamma^e{}_{ac}(\partial_b\omega_e - \Gamma^d{}_{be}\omega_d)\right] \\
&\quad - \left[\partial_b(\partial_a\omega_c - \Gamma^d{}_{ac}\omega_d) - \Gamma^e{}_{ba}(\partial_e\omega_c - \Gamma^d{}_{ec}\omega_d) - \Gamma^e{}_{bc}(\partial_a\omega_e - \Gamma^d{}_{ae}\omega_d)\right] \\
&= -(\partial_a\Gamma^d{}_{bc})\omega_d + (\partial_b\Gamma^d{}_{ac})\omega_d + \Gamma^e{}_{ac}\Gamma^d{}_{be}\omega_d - \Gamma^e{}_{bc}\Gamma^d{}_{ae}\omega_d \\
&= \left[-2\partial_{[a}\Gamma^d{}_{b]c} + 2\Gamma^e{}_{c[a}\Gamma^d{}_{b]e}\right]\omega_d. \qquad (6)
\end{aligned}
$$

where the third equality uses the commutativity of ordinary derivatives, and the fourth equality uses the symmetry of $\Gamma^c{}_{ab}$.Since this holds for arbitrary ω_d, we obtain

$$
R_{abc}{}^d = -2\partial_{[a}\Gamma^d{}_{b]c} + 2\Gamma^e{}_{c[a}\Gamma^d{}_{b]e}. \qquad (7)
$$

Writing this in component form, we obtain

$$
R_{\mu\nu\rho}{}^\sigma = \frac{\partial}{\partial x^\nu}\Gamma^\sigma{}_{\mu\rho} - \frac{\partial}{\partial x^\mu}\Gamma^\sigma{}_{\nu\rho} + \left(\Gamma^\alpha{}_{\mu\rho}\Gamma^\sigma{}_{\alpha\nu} - \Gamma^\alpha{}_{\nu\rho}\Gamma^\sigma{}_{\alpha\mu}\right), \qquad (8)
$$

where we have used the definition of the ordinary derivative as the partial derivative of components with respect to the coordinates.

2 The Ricci Tensor

From Eqs. (8) and (2), we have

$$
R_{\mu\rho} = \frac{\partial}{\partial x^\nu}\Gamma^\nu{}_{\mu\rho} - \frac{\partial}{\partial x^\mu}\Gamma^\nu{}_{\nu\rho} + \left(\Gamma^\alpha{}_{\mu\rho}\Gamma^\nu{}_{\alpha\nu} - \Gamma^\alpha{}_{\nu\rho}\Gamma^\nu{}_{\alpha\mu}\right), \qquad (9)
$$

Eq. (9) contains the "contracted Christoffel symbol" $\Gamma^\nu{}_{\nu\sigma}$, whose expression we now derive. From

$$
\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2}g^{\sigma\rho}(g_{\rho\mu,\nu} + g_{\nu\rho,\mu} - g_{\mu\nu,\rho}), \qquad (10)
$$

we obtain

$$
\Gamma^\mu{}_{\mu\sigma} = \frac{1}{2}g^{\mu\lambda}(g_{\sigma\lambda,\mu} + g_{\mu\lambda,\sigma} - g_{\mu\sigma,\lambda}) = \left(\frac{1}{2}g^{\mu\lambda}g_{\mu\lambda,\sigma} + g^{\mu\lambda}g_{\sigma[\lambda,\mu]}\right) = \frac{1}{2}g^{\mu\lambda}g_{\mu\lambda,\sigma}, \qquad (11)
$$

where in the last step we used $g^{[\mu\lambda]} = 0$. Rewriting the above as

$$
\Gamma^\mu{}_{\mu\sigma} = \frac{1}{2}g^{\mu\lambda}\frac{\partial g_{\mu\lambda}}{\partial x^\sigma}. \qquad (12)
$$

On the other hand, the determinant g of the matrix formed by $g_{\mu\lambda}$ can be expanded along the μ-th row as $g = g_{\mu\lambda}A^{\mu\lambda}$ (summing only over λ, where $A^{\mu\lambda}$ is the cofactor of $g_{\mu\lambda}$), so $\partial g/\partial g_{\mu\lambda} = A^{\mu\lambda}$. From the inverse matrix relation

$$
g_{\mu\lambda}g^{\lambda\nu} = \delta^\nu_\mu, \qquad (13)
$$

treating $g^{\lambda\nu}$ as the unknown for fixed ν, Cramer's rule gives

$$
g^{\lambda\nu} = \frac{A^{\nu\lambda}}{g}, \qquad (14)
$$

where $A^{\nu\lambda}$ is the cofactor of $g_{\nu\lambda}$. Hence

$$\frac{\partial g}{\partial g_{\mu\lambda}} = A^{\mu\lambda} = g g^{\lambda\mu} = g g^{\mu\lambda}. \tag{15}$$

Since $g_{\mu\lambda}$ are functions of the coordinates x^σ, so is g, and

$$\frac{\partial g}{\partial x^\sigma} = \frac{\partial g}{\partial g_{\mu\lambda}} \frac{\partial g_{\mu\lambda}}{\partial x^\sigma} = g\, g^{\mu\lambda} \frac{\partial g_{\mu\lambda}}{\partial x^\sigma}, \tag{16}$$

where in the last step we used Eq. (15). Combining Eqs. (12) and (16) gives

$$\Gamma^\mu{}_{\mu\sigma} = \frac{1}{2g} \frac{\partial g}{\partial x^\sigma} = \frac{1}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\sigma}. \tag{17}$$

Substituting Eq. (17) into Eq. (9), we obtain the Ricci tensor.

3 Other Applications of the Contracted Christoffel Symbol

Using the coordinate basis, from Eq. (17) and $\nabla_a v^a = \partial_a v^a + \Gamma^a{}_{ab} v^b$, we readily obtain

$$\begin{aligned}
\nabla_a v^a &= \partial_\mu v^\mu + \Gamma^\mu{}_{\mu\nu} v^\nu \\
&= \partial_\mu v^\mu + \frac{1}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\nu} v^\nu \\
&= \frac{1}{\sqrt{|g|}} \sqrt{|g|} \partial_\sigma v^\sigma + \frac{1}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\sigma} v^\sigma \\
&= \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial x^\sigma} \left(\sqrt{|g|}\, v^\sigma \right).
\end{aligned} \tag{18}$$

This formula can be used to conveniently compute divergences.

References

[1] Wald, R. M. *General Relativity*. Chicago: The University of Chicago Press, 1984.

[2] Liang, C. B. and Zhou, B. *Introduction to Differential Geometry and General Relativity, Volume I*. 2nd ed. Beijing: Science Press, 2006.

[3] DeepSeek. *DeepSeek* [Large AI Model]. 2024. URL: `https://www.deepseek.com` (Accessed: July 28, 2026).

A Appendix

A.1 Report on Use of AI Tools

I primarily use AI for simple repetitive calculations, as well as for English translation and polishing. AI also assists me in explaining concepts, verifying my ideas, and providing valuable proof strategies.

A.2 What Is New in This Work

This text is my reading notes on Wald's book [1], intended to supplement the specific computational details, while following the basic content of the original work.

Specifically, in Sec. 1, I provide a detailed computation of Eq. (6). In Sec. 2, I add some computational details. In Sec. 3, I supply the proof of the expression for the divergence of a vector field.

Finally, it should be emphasized that this note is merely a set of reading notes on Wald's book; the overall logic and ideas are inherited from Wald's work. My contribution consists only of rephrasing, reorganizing, and supplementing the specific computational and proof details. In addition, Sec. 2 draws heavily on the work of Liang and Zhou [2]. If there is anything of value in this note, the credit should first go to Wald's book and Liang's book. Should the copyright holder believe that this note infringes upon the copyright of either work, please contact me and I will remove it promptly.